

DUNKERQUE, FranceWMO: **070100**Lat: **51.0558N**Lon: **2.3394E**Elev: **16**StdP: **101.13**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.2	-1.5	-7.3	2.0	-0.7	-5.4	2.4	-0.5	15.9	7.7	14.6	7.4	5.6	80	0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.4	25.2	18.9	23.4	18.4	22.0	17.8	20.2	23.9	19.3	22.3	18.5	21.2	4.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.6	22.1	18.0	13.0	21.2	17.3	12.4	20.4	57.9	24.0	55.1	22.4	52.5	21.3	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	12.1	10.9	DB	-4.5	32.1	1.8	3.1	-5.8	34.4	-6.9	36.2	-7.9	38.0	-9.3	40.2	
(5)			WB	-5.5	22.0	2.4	0.8	-7.2	22.6	-8.6	23.0	-10.0	23.5	-11.7	24.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	5.5	5.8	7.6	10.2	13.2	16.0	18.2	18.6	16.3	12.9	9.1	6.2
	DBStd	5.45	3.63	3.29	2.74	2.75	2.62	2.39	2.17	2.00	2.18	2.84	2.98	3.57
	HDD10.0	556	141	119	82	29	3	0	0	0	8	49	123	123
	HDD18.3	2498	397	351	332	244	160	78	28	19	68	169	277	376
	CDD10.0	1169	3	3	8	36	103	179	254	267	189	99	23	6
	CDD18.3	69	0	0	0	0	2	8	24	28	7	1	0	0
	CDH23.3	188	0	0	0	1	10	33	56	68	19	2	0	0
	CDH26.7	47	0	0	0	0	2	9	15	16	4	0	0	0
(14) Wind	WSAvg	5.7	6.5	6.4	6.0	5.5	5.5	5.3	5.2	5.1	5.4	5.6	5.8	6.3
(15) Precipitation	PrecAvg	685	58	39	53	45	50	52	54	51	64	73	85	61
	PrecMax	866	136	81	113	129	156	125	138	100	141	201	194	124
	PrecMin	508	11	4	8	6	9	0	10	6	5	12	35	18
	PrecStd	105	27	21	28	25	33	30	30	25	39	46	42	31
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	14.4	17.7	21.4	24.3	27.7	28.3	29.4	25.5	22.3	16.7	14.3
		MCWB	11.4	10.3	12.5	13.9	16.1	19.0	19.6	20.5	18.2	17.1	14.3	12.3
	2%	DB	12.4	12.2	13.9	17.8	20.3	23.0	24.4	25.1	22.6	19.2	15.3	13.3
		MCWB	10.8	9.9	10.7	12.5	15.0	18.1	19.1	19.3	17.7	16.1	13.4	11.7
	5%	DB	11.1	11.1	12.2	15.5	18.3	20.9	22.8	23.1	20.8	17.7	14.3	12.2
		MCWB	9.6	9.3	9.9	11.7	14.4	16.9	18.4	18.9	17.0	15.2	12.6	10.7
	10%	DB	10.1	10.2	11.1	13.8	16.8	19.4	21.5	21.7	19.5	16.6	13.1	11.1
		MCWB	8.6	8.5	9.2	10.8	13.6	15.9	17.7	18.0	16.3	14.4	11.5	9.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	11.5	12.7	14.4	17.2	20.1	21.3	21.5	19.9	17.6	14.7	12.9
		MCDB	12.8	13.4	15.8	19.6	21.6	25.8	26.0	27.0	23.5	20.6	15.8	13.9
	2%	WB	10.9	10.4	11.2	13.1	15.7	18.4	20.1	20.3	18.6	16.5	13.7	11.9
		MCDB	12.2	11.8	13.3	16.8	19.1	22.0	23.6	23.8	21.2	18.8	15.0	13.1
	5%	WB	9.8	9.5	10.3	12.1	14.8	17.3	19.1	19.5	17.7	15.6	12.8	10.9
		MCDB	11.0	10.9	11.9	14.9	17.8	20.2	22.0	22.3	19.9	17.3	14.1	12.2
	10%	WB	8.7	8.7	9.5	11.2	14.0	16.4	18.3	18.6	16.9	14.8	11.7	9.7
		MCDB	9.9	10.1	11.0	13.5	16.4	19.0	20.8	21.2	19.1	16.4	13.0	10.9
(35) Mean Daily Temperature Range	5% DB	MDBR	3.7	4.0	4.4	5.2	5.1	5.2	5.3	5.4	5.2	4.9	3.9	3.8
		MCDBR	4.4	5.2	6.4	8.6	8.0	8.3	7.8	7.8	7.8	6.3	4.4	4.3
	5% WB	MCWBR	4.1	3.9	4.2	4.6	4.3	4.7	4.2	4.2	4.1	4.1	3.8	4.2
		MCDBR	4.3	4.6	5.6	7.6	7.1	7.5	6.8	6.8	6.5	5.7	4.2	4.3
		MCWBR	4.3	3.9	4.1	4.3	4.1	4.6	4.1	3.9	4.1	4.2	3.9	4.2
(40) Clear-Sky Solar Irradiance	taub	0.335	0.354	0.393	0.410	0.420	0.419	0.418	0.409	0.388	0.371	0.346	0.329	0.329
	taud	2.366	2.348	2.243	2.227	2.255	2.279	2.290	2.333	2.380	2.395	2.380	2.350	2.350
	Ebn at Noon	676	764	796	827	836	839	833	821	797	734	654	616	616
	Edh at Noon	65	84	112	127	129	127	124	113	97	80	63	57	57
(44) All-Sky Solar Radiation	RadAvg	0.79	1.48	2.69	4.36	5.11	5.57	5.28	4.49	3.34	1.92	0.92	0.61	0.61
	RadStd	0.07	0.27	0.34	0.41	0.39	0.46	0.52	0.38	0.23	0.17	0.10	0.08	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (71 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A